

04/10/2026

The six functors

and convolution

Last time Presentability, started background on the six functors

Today More on the six functors, convolution products

Recollement

Prop. Let X be a space, $i: Z \hookrightarrow X$ a closed subspace, and $j: U = X \setminus Z \hookrightarrow X$ its open complement. Then:

(1) $i_*: \mathrm{Sh}(Z, \mathcal{A}_n) \longrightarrow \mathrm{Sh}(X, \mathcal{A}_n)$ is fully faithful.

(2) $i^*: \mathrm{Sh}(X, \mathcal{A}_n) \longrightarrow \mathrm{Sh}(Z, \mathcal{A}_n)$ and $j^*: \mathrm{Sh}(X, \mathcal{A}_n) \longrightarrow \mathrm{Sh}(U, \mathcal{A}_n)$ are jointly conservative.

Write $\mathrm{Glue}(Z, U)$ for the ∞ -category of triples

$$(F_Z \in \mathrm{Sh}(Z), F_U \in \mathrm{Sh}(U), \phi: F_Z \longrightarrow i^* j_*(F_U)).$$

More precisely, $\mathrm{Glue}(Z, U)$ is the pullback in Cat_∞

$$\begin{array}{ccc} \mathrm{Glue}(Z, U) & \longrightarrow & \mathrm{Fun}([1], \mathrm{Sh}(Z, \mathcal{A}_n)) \\ \downarrow \lrcorner & & \downarrow \text{target} \\ \mathrm{Sh}(U, \mathcal{A}_n) & \xrightarrow{i^* j_*} & \mathrm{Sh}(Z, \mathcal{A}_n) \end{array}$$

(3) The functor

$$\mathrm{Sh}(X, \mathcal{A}_n) \longrightarrow \mathrm{Glue}(Z, U)$$

$$F \longmapsto (i^*(F), j^*(F), i^*(F) \xrightarrow{i^*(\mathrm{unit})} i^*j_*j^*(F))$$

is an equivalence. The inverse sends $(F_Z, F_U, \phi: F_Z \rightarrow i^*j_*(F_U))$ to the pullback of

$$i_*(F_Z) \xrightarrow{i_*(\phi)} i_*i^*j_*(F_U) \xleftarrow{\mathrm{unit} \circ j_*} j_*(F_U).$$

Let $\mathcal{E}$ be a presentable ∞ -category. Then (1) is true with $\mathcal{A}_n$ replaced by $\mathcal{E}$. If $\mathcal{E}$ is compactly generated or stable, then (2) and (3) are true with $\mathcal{A}_n$ replaced by $\mathcal{E}$.

Remark. In this situation, $j_!$ is uniquely characterized by

$$j^*j_! \simeq \mathrm{id} \quad \text{and} \quad i^*j_* \simeq \emptyset \leftarrow \text{constant w/ value the initial object}$$

The exceptional functors

Def. A map of locally compact Hausdorff spaces $f: X \rightarrow Y$ is proper if for every compact subspace $K \subset Y$, the preimage $f^{-1}(K)$ is compact.

Fact. Proper maps are stable under composition and basechange.

Thm (Lurie). If $f: X \rightarrow Y$ is a proper map of locally compact Hausdorff spaces, then

$$f_* \operatorname{Sh}(X, \mathcal{A}_n) \longrightarrow \operatorname{Sh}(Y, \mathcal{A}_n)$$

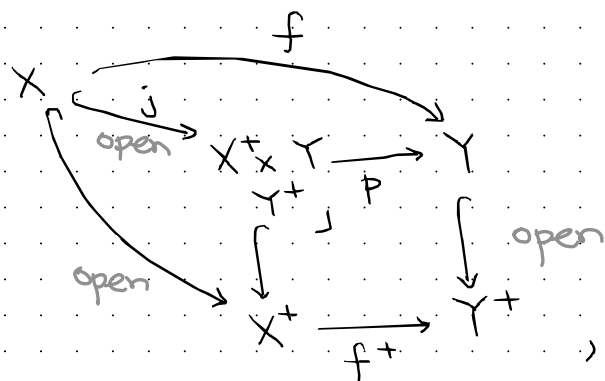
preserves filtered colimits. As a consequence, for any presentable stable ∞ -category $\mathcal{E}$, the functor

$$f_* \operatorname{Sh}(X, \mathcal{E}) \longrightarrow \operatorname{Sh}(Y, \mathcal{E})$$

admits a right adjoint $f^!$.

Ntn Write $\text{LCH} \subset \text{Top}$ for the full subcategory spanned by the locally compact Hausdorff spaces. The subcategory $\text{LCH} \subset \text{Top}$ is closed under finite limits.

Observe Every map $f: X \rightarrow Y$ admits a factorization $f = p \circ j$ where j is an open immersion and p is proper. For example, take



where $(-)^+$ denotes one-point compactification.

Idea: We'd like to define an adjunction

$$\text{Sh}(X; \mathcal{E}) \begin{matrix} \xrightarrow{f_!} \\ \perp \\ \xleftarrow{f^!} \end{matrix} \text{Sh}(Y; \mathcal{E})$$

by setting $f_! := p_* j_!$

and $f^! := j^* p^!$

- > But it isn't obvious this is independent of the choice of factorization and can be made functorial.
- > Nonetheless, it is true. We'll soon state this. The key inputs are basechange theorems.

Basechange

Thm Given a pullback square of

$$\begin{array}{ccc} W & \xrightarrow{\bar{f}} & Y \\ \bar{g} \downarrow \lrcorner & & \downarrow g \\ X & \xrightarrow{f} & Z \end{array} \quad \text{of LCH spaces}$$

for sheaves valued in a presentable ∞ -cat that is compactly gen or stable, we have:

(open basechange) If f is an open immersion, the natural map $\bar{f}_! \bar{g}^* \longrightarrow g^* f_!$

is an isomorphism

(proper basechange) If g is proper, the natural map

$$f^* g_! = f^* g_* \longrightarrow \bar{g}_* \bar{f}^* = \bar{g}_! \bar{f}^*$$

is an isomorphism

$\otimes$ and Hom

Observe If $\mathcal{C}$ is a presentably Sym mon ∞ -category, then the adjoint functor theorem guarantees that the Sym mon structure is closed, so there is a functor

$$\text{Hom}_{\mathcal{C}} : \mathcal{C}^{\text{op}} \times \mathcal{C} \longrightarrow \mathcal{C}$$

and natural isos

$$\text{Map}_{\mathcal{C}}(X \otimes Y, Z) \simeq \text{Map}_{\mathcal{C}}(X, \text{Hom}_{\mathcal{C}}(Y, Z))$$

Convention From now on, let's work with sheaves of Spectra.

Ntn Write $\mathbb{1}_X$ or $\mathbb{1}$ in $\text{Sh}(X) = \text{Sh}(X, \text{Sp})$ for the constant sheaf of Spectra with value the Sphere Spectrum.

Thm. For any top Space X , the stable ∞ -category $\mathrm{Sh}(X)$ has a unique presentably symmetric monoidal structure $\otimes$ with the following properties:

(1) 1_X is the unit

(2) Given opens $j: U \hookrightarrow X$ and $j': U' \hookrightarrow X$ with inters $j'': U \sqcup U' \hookrightarrow X$, there is a natural iso.

$$j_!(1_U) \otimes j'_!(1_{U'}) \xrightarrow{\sim} j''_!(1_{U \sqcup U'})$$

Categorical Künneth formula

Thm (Lurie). Let X and Y be locally compact Hausdorff spaces. Then the functor

$$\mathrm{Sh}(X) \times \mathrm{Sh}(Y) \longrightarrow \mathrm{Sh}(X \times Y)$$

$$(F, G) \longmapsto \mathrm{pr}_1^*(F) \otimes \mathrm{pr}_2^*(G)$$

induces an equivalence

$$\mathrm{Sh}(X) \otimes \mathrm{Sh}(Y) \xrightarrow{\sim} \mathrm{Sh}(X \times Y).$$

↑
tensor product
of presentable
 ∞ -categories

The Six-functor formalism

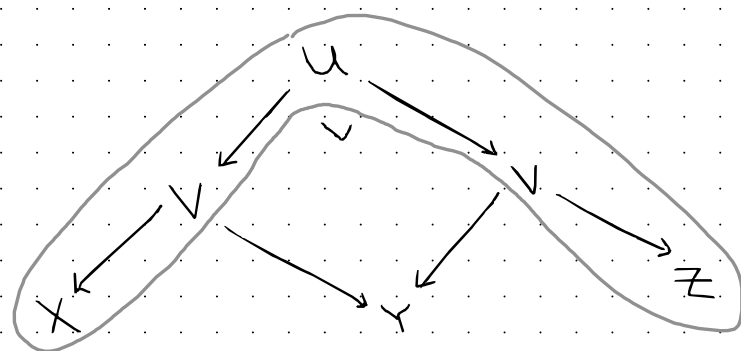
Notation Write $\text{Corr}(\text{LCH})$ or $\text{Span}(\text{LCH})$ for the 2-category with objects locally compact Hausdorff spaces and

$$\text{Map}(X, Y) = \left\{ \begin{array}{c} \text{groupoid of spans} \\ \begin{array}{ccc} & V & \\ f \swarrow & & \searrow g \\ X & & Y \end{array} \end{array} \right\}$$

> Composition is given by pullback

$$\left(\begin{array}{ccc} & W & \\ \swarrow & & \searrow \\ Y & & Z \end{array} \right) \circ \left(\begin{array}{ccc} & V & \\ \swarrow & & \searrow \\ X & & Y \end{array} \right)$$

ii.



Note $\text{Corr}(\text{LCH})$ has a symmetric monoidal structure given by cartesian product (which is **not** the product in this category):

$$X \otimes X' := X \times X' \text{ in } \text{LCH}.$$

Note There is a faithful functor

$$\text{Lleg } \text{LCH}^{\text{op}} \longrightarrow \text{Corr}(\text{LCH})$$

$$[X \xrightarrow{f} Y] \longmapsto \left[Y \xleftarrow{f} X \right]_{=X}$$

Thm (Lurie, Volpe, Mann, ...) The functor $\text{Sh} : \text{LCH}^{\text{op}} \rightarrow \text{Pr}^{\text{L}}$ with $*$ -pullback functoriality has a canonical extension to a symmetric monoidal functor

$$(\text{Corr}(\text{LCH}), \otimes) \longrightarrow (\text{Pr}^{\text{L}}, \otimes)$$

$$\begin{array}{ccc} X & \xleftarrow{f} & V \\ & & \searrow g \\ & & Y \end{array} \longmapsto g_! f^* : \text{Sh}(X) \longrightarrow \text{Sh}(Y)$$

Here, extension means that

$$\mathbf{LCH}^{\text{op}} \xrightarrow{\text{Lleg}} \mathbf{Corr}(\mathbf{LCH}) \longrightarrow \mathbf{Pr}^{\mathbf{L}}$$

is Sh. Moreover

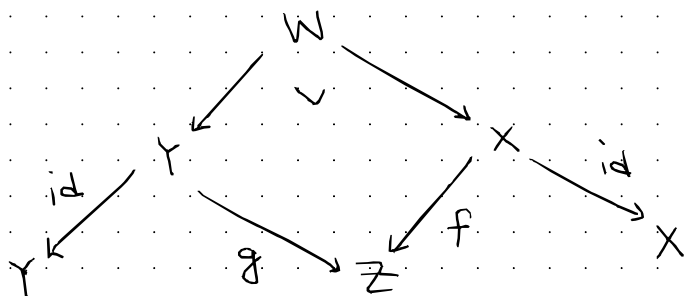
- (1) If $f = \text{id}$ and g is an open immersion, then $g_! f^* = g_!$ as previously defined (the left adjoint to g^*)
- (2) If $f = \text{id}$ and g is proper, then $g_! f^* = g_*$ as previously defined.

Note

- (1) Letting f or g be the identity provides $(-)^*$ or $(-)_!$ functionality.
- (2) The functor $\text{Lleg} : (\mathbf{LCH}^{\text{op}}, x) \longrightarrow (\mathbf{Corr}(\mathbf{LCH}), \otimes)$ is sym. mon.
 - Since every object of an ∞ -category with finite products is naturally a commutative coalgebra (with comult the diagonal $\Delta: x \rightarrow x \times x$), every object of $\mathbf{Corr}(\mathbf{LCH})$ is naturally a commutative algebra.

- Since (lax) symmetric monoidal functors preserve comm algebras, each $\mathrm{Sh}(X)$ is a commutative algebra in $(\mathrm{Pr}_{\mathrm{L}}^{\mathrm{L}}, \otimes)$, i.e., a presentably symmetric monoidal ∞ -category and each f^* is symmetric monoidal.
- This symmetric monoidal structure agrees with the previously-mentioned one.

(3) Compatibility of $g_! f^*$ with Composition $\iff$ basechange formulas.



says that

$$f^* g_! \simeq \bar{g}_! \bar{f}^*$$

as functors $\mathrm{Sh}(Y) \rightarrow \mathrm{Sh}(X)$

Remark Given $f: X \rightarrow Y$ and $E \in \text{Sh}(Y)$, the following are the sheaf-theoretic interpretations of E -(co)homology of X over Y :

Cohomology $f_* f^*(E)$

Borel-Moore homology (or bivariant homology) $f_* f^!(E)$

Compactly Supported Cohomology $f_! f^*(E)$

Homology $f_! f^!(E)$

Duality

Def. For a map $f: X \rightarrow Y$ in LCH, the dualizing sheaf is

$$\omega_f = f^!(\mathbb{1}_Y)$$

> If f is proper, then $f_* \dashv f^!$, so

$$\mathrm{Hom}(F, f^!(\mathbb{1})) \simeq \mathrm{Hom}(f_*(F), \mathbb{1}) \leftarrow \text{The functional dual of } f_*(F)$$

Thm (relative Poincaré duality) Let $f: X \rightarrow Y$ be a C^1 -submersion between C^1 -manifolds. Then

(1) $f^!(\mathbb{1}) \in \mathrm{Sh}(X)$ is locally constant and $\otimes$ -invertible

(2) There is a natural isomorphism

$$f^!(\mathbb{1}) \otimes f^*(g) \xrightarrow{\sim} f^!(g)$$

(3) There is an explicit formula for $f^!(1)$ in terms of the relative tangent bundle of f . What we'll need is that if f is a projection $Y \times \mathbb{R}^n \rightarrow Y$, then $f^!(1) \simeq 1[n]$ and

$$f^! \simeq f^*[n]$$

The convolution product

Observe. The functor

$$\mathbf{Rleg}(\mathbf{LCH}, \times) \longrightarrow (\mathbf{Corr}(\mathbf{LCH}), \otimes)$$

$$[f: X \rightarrow Y] \longmapsto \left[\begin{array}{c} X \\ \parallel \\ X \end{array} \xrightarrow{f} Y \right]$$

is symmetric monoidal, hence induces a functor

$$\mathbf{Rleg} \mathbf{CMon}(\mathbf{LCH}) \longrightarrow \mathbf{CAlg}(\mathbf{Corr}(\mathbf{LCH}))$$

- > Applying $\mathbf{Sh}$, we deduce that if $M \in \mathbf{CMon}(\mathbf{LCH})$ with addition $a: M \times M \rightarrow M$ and zero $0: * \rightarrow M$, then $\mathbf{Sh}(M)$ acquires a presentably symmetric monoidal structure $*$ given by

$$F * G = a_! (pr_1^*(F) \otimes pr_2^*(G)).$$

The unit for convolution is $0_!(1) = 0_*(1)$, i.e., the skyscraper

sheaf at 0 with value the unit

Def. We call this symmetric monoidal structure on $\text{Sh}(M)$ the convolution product

Observe If $M = 0$ is the zero comm. monoid, then the convolution product on $\text{Sh}(M) = \text{Sp}$ agrees with the usual tensor product.

Observe If $\phi: M \rightarrow N$ is a map in $\text{CMon}(\text{LCH})$, then

$$\phi_!: \text{Sh}(M) \rightarrow \text{Sh}(N)$$

is symmetric monoidal for the convolution product, Hence its right adjoint $\phi^!: \text{Sh}(N) \rightarrow \text{Sh}(M)$ is lax symmetric monoidal for convolution.

Observe So if $M \in \text{CMon}(\text{LCH})$ and $p: M \rightarrow 0$ is the unique map, the dualizing sheaf $\omega_M := p^!(1)$ acquires the struct. of a commutative algebra for convolution.